

A MAJOR PROJECT REPORT
ON
ENHANCEMENT OF VIRTUAL ASSISTANT THROUHH
MULTIMODAL AI FOR EMOTION RECOGNITION

Submitted to

SRI INDU COLLEGE OF ENGINEERING AND TECHNOLOGY

In partial fulfillment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY

In

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

SRI INDU COLLEGE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institution under UGC, accredited by NBA, Affiliated to JNTUH)

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(2024-2025)

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



CERTIFICATE

Certified that the Major Project entitled “**ACCESS CONTROL BY SIGNATURE KEYS TO PROVIDE PRIVACY FOR CLOUD AND BIG DATA**” is a Bonafide work carried out by **D. KAVITHA SRI(21D41A7214), B. SPANATH(21D41A7204), B. AJAY (21D41A7203)** in partial fulfillment for the award of **Bachelor of Technology** in **ARTIFICIAL INTELLIGENCE AND DATA SCIENCE** of **SICET, Hyderabad** for the academic year **2024- 2025**. The Project has been approved as it satisfies academic requirements in respect of the work prescribed for **IV YEAR, II-SEMESTER** of **B. TECH** course.

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ACKNOWLEDGEMENT

With great pleasure we want to take this opportunity to express our heartfelt gratitude to all the people who helped in making this project work a success. We thank the almighty for giving us the courage & perseverance in completing the project.

We are thankful to principal **Prof. Dr. G. Suresh**, for giving us the permission to carry out this project and for providing necessary infrastructure and labs.

We are highly indebted to, **Dr. T. Charan Singh**, Head of the Department of Artificial Intelligence and Data Science, Project Guide, for providing valuable guidance at every stage of this project.

We are grateful to our internal project guide, Mr. **R. MADHAVA REDDY** for his constant motivation and guidance given by him during the execution of this project work.

We would like to thank the Teaching & Non-Teaching staff of Department of Artificial Intelligence and Data Science for sharing their knowledge with us.

Last but not the least we express our sincere thanks to everyone who helped directly or indirectly for the completion of this project.

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1. INTRODUCTION

Cloud computing and Big data as novel techniques need more attention and research. The privacy for these novel techniques is one of the most important issues. Shared data or processing and transferring data in third party could be more vulnerable to attacks, such as sync cookies, attacks on client profiles, limited connections of provided, etc. [1]. Cloud computing is seen as an essential, low-maintenance way to share resources. More and more, moving the systems that manage the local information into cloud servers has become the standard procedure, clients being able to benefit of premium services whilst saving big money on the local infrastructures. Users that use cloud computing are no longer faced with the disadvantages of the problematic local solutions for storing and management of data. Using policy of encrypted data based on access control is the most common privacy method. This policy can ensure privacy of data that represent sensitive information.

The privacy based on access control means to allow access to data only to authorized persons. The access mechanisms to the sensitive data have problems if they can be shared without strong privacy. Data is often in cloud computing with shared access with third party, which makes it more vulnerable to attacks. Usually, moving data between sides can be risky on client privacy. To ensure end-to-end security, we try to implement algorithms to provide strong privacy for big data in cloud. Other related works and research in same area, as the encryption based-attribute, show the most suitable approach to identify efficient and more scalable methods. Cloud computing implies a set of computers that are used together to provide different accounts and services. The benefits of using cloud computing in companies are cost reduction and time saving. Also, using shared services from cloud is easier than building and developing own infrastructure. The providers of cloud computing are focus on providing flexible services, cost-effective IT infrastructure and secure environments for companies and organizations [4].

The main issue with big data in cloud is that processing or usage always needs to be done by third party. It is very important for the owners of data, or clients, to trust and to have the guarantee of privacy for the information stored in cloud or analyzed as big data. The privacy methods studied in previous research showed that privacy infringement for cloud computing and big data happened because of limitation, privacy guarantee rate or risks on data by external or internal attackers. Generally, the private client data has been under attack.

2. LITERATURE SURVEY

TITLE: A Privacy Preserving System for Cloud Computing.

AUTHOR: Ulrich xzzzq, Benjamin Justus, Dennis Loehr,(2011),

ABSTRACT: Cloud computing is changing the way that organizations manage their data, due to its robustness, low cost and ubiquitous nature. Privacy concerns arise whenever sensitive data is outsourced to the cloud. This paper introduces a cloud database storage architecture that prevents the local administrator as well as the cloud administrator to learn about the outsourced database content. Moreover, machine readable rights expressions are used in order to limit users of the database to a need-to-know basis. These limitations are not changeable by administrators after the database related application is launched, since a new role of rights editors is defined once an application is launched. Furthermore, trusted computing is applied to bind cryptographic key information to trusted states. By limiting the necessary trust in both corporate as well as external administrators and service providers, we counteract the often criticized privacy and confidentiality risks of corporate cloud computing.

TITLE: Enhance Big Data Security in Cloud Using Access Control.

AUTHOR: Yong Wang, Ping Zhang, (2017),

ABSTRACT: Big data has become an essential requirement for enterprises looking to harness their business potential. The use cases for big data are endless and range from customer targeting and fraud analytics to anomaly detection and more. This data can be generated quickly from various sources such as users' browser and search history, credit card payments, mobile pinging of the nearest cell phone tower, etc. Given the volume of sensitive information being captured, any or accidental disclosure of or access to the data can have severe consequences for your enterprise, both in financial terms and in more intangible ways, such as the loss of brand recognition and users' trust.

In recent years, many highly scalable and complex processing *frameworks for big data* have emerged, such as Hadoop, Hive, Presto, and Spark. Securing these frameworks is very challenging because of their distributed nature, and involves many touchpoints, services, and operational processes. With accelerating cloud adoption, monitoring access and data flows for big data becomes even more complicated.

TITLE: Security Techniques for Data Protection in Cloud Computing.

AUTHOR: Kire Jakimoski, (2016),

ABSTRACT: The IBM Security® Guardium® Insights data security platform allows any enterprise to quickly address their data security and compliance needs. Its robust capabilities help enterprises automate compliance policy enforcement and centralize data activity from multiple clouds. This enhances visibility to achieve a consolidated view of how critical data is being accessed and used across hybrid environments. Guardium Insights is designed to provide data security specialists with a centralized hub where they can retain monitoring data for compliance or analytic purposes for as long as they need to. With best-in-class features such as automated compliance, audit and reporting, and real-time monitoring, Guardium Insights can help users achieve a reduction in audit prep time. Guardium Insights can complement and enhance existing Guardium® Data Protection deployments or be installed on its own to help solve compliance and cloud data activity monitoring challenges.

TITLE: Toward cloud-based collaboration services

AUTHOR: Banks, David, John S. Erickson, and Michael Rhodes. (2009)

ABSTRACT: As we approach the end of the first decade of the 21st century, we are witnessing a disruptive change in the provisioning of information technology: the advent of cloud computing. For most organizations, information technology is not a core competence. Until recently, their only option was to retain IT specialists on premise, but now alternatives from the likes of Google, Amazon and Salesforce.com are becoming increasingly viable. Accordingly, out-sourced IT is now an option for all sizes of company. This coincides with businesses seeking to operate efficiently in a global marketplace by outsourcing noncore competencies. As businesses choose to excel in a single area and partner for the rest, collaboration across organizational boundaries becomes a core part of product development. Traditional Enterprise Content Management (ECM) software has not kept up, leaving people collaborating via email—the lowest common denominator. In response to these trends, we envision a generation of cloud-based collaboration platforms emerging to address the needs of content-centered collaboration between businesses, millions of users. The Fractal research program [1] in HP Labs aims to design and deploy such a platform.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Cloud computing is seen as an essential, low-maintenance way to share resources. More and more, moving the systems that manage the local information into cloud servers has become the standard procedure, clients being able to benefit of premium services whilst saving big money on the local infrastructures. Users that use cloud computing are no longer faced with the disadvantages of the problematic local solutions for storing and management of data. Using policy of encrypted data based on access control is the most common privacy method. This policy can ensure privacy of data that represent sensitive information. The privacy based on access control means to allow access to data only to authorized persons. The access mechanisms to the sensitive data have problems if they can be shared without strong privacy.

DISADVANTAGES

1. Data is often in cloud or big data with shared access with third party, which makes it more vulnerable to attacks. Usually, moving data between sides can be risky on client privacy.
Key Distribution: Distributing signature keys securely to authorized users or entities can be challenging. If not handled properly, it can lead to key exposure or unauthorized access. Implementing secure key distribution mechanisms is crucial
Human Error: Users may accidentally mishandle their signature keys, leading to access problems or security vulnerabilities. Training and user education are essential to mitigate this.